

1. Healthcare and IoT Device Vulnerability Testbed**2. Team Members**

- Justin Bower - jbower2024@my.fit.edu
- Nathan Maloney - nmaloney2022@my.fit.edu
- Ipule Pipi - ipipi2022@my.fit.edu
- Jordan Chesley - jchesley2022@my.fit.edu

3. Faculty advisor(s): Dr. Sneha Sudhakaran and Dr. Abdullah Aydeger**4. Client(s):** Security Researchers/Manufacturers/Engineers**5. Progress of Current Milestone (Milestone 4 - September 28th)**

Task	% Complete	Justin	Nathan	Ipule	Jordan	To-Do
Project Management, Document Creation, and Overall Planning	100%	100%	0%	0%	0%	Continue managing project and creating documents as planning is now complete
Complete Firmware and Adafruit Demos	100%	33%	33%	33%	0%	Web interface demo of database functions
VR/RE & CVE database work	75%	50%	0%	0%	25%	Finish VR/RE of smart watches/heart monitors and enter any CVES
Local GUI development	50%	25%	0%	0%	25%	Add launch options, complete frame-capture for tooling, and integrate script launching
Testbed and Networking Set-up	90%	30%	0%	30%	30%	Flash the router with OpenWRT, other than that the testbed/lab is complete
Web Dashboard & Cloud Backend Integration for	0%	0%	60%	40%	0%	Integrate analysis initiation, job status, results/artifacts, workflow states, and authorization into dashboard
Cloud infrastructure & backend services	90%	0%	90%	0%	0%	Cloud API endpoints and firewall for local application
Artifact & Firmware Management	100%	0%	50%	50%	0%	Continue integration with analysis workflow

6. Discussion of each accomplished task (and obstacles) for Milestone 4:

• Task 1:

Up until this point project, Justin alone has been completing approximately 60-90 percent of documents, including during semester 1. This is an obstacle; however, the rest of the team contributed much more to this milestone document. In addition, Jordan has joined the project which will alleviate this document overhead issue, and he can assist Justin with RE and local GUI development. Prior to this point, Justin was the only one in the project with VR/RE experience.

• Task 2:

Our demos for uploading firmware/binary file artifacts to the web interface are complete. In addition, the demo for using the Adafruit sniffer and the open source SnifferAPI for pulling pcap files from devices is complete. The only obstacle is ensuring that the SnifferAPI works in our NixOS environment, as the demo is in a debian environment, however NixOS is based on linux so it is just a matter of modifying the NixOS configuration file

• Task 3:

Upon continuing the research work from last semester, we wanted to begin getting some more firmware artifacts, and utilize the tooling we now had in semester 2. However, we quickly discovered that the firmware of the inexpensive and generic wifi chipsets is typically not stored on device, hence the lack of UART/JTAG connections. Rather than using an EEPROM or PCB chip reader, the solution was to write a portion of the NixOS config, to pull the firmware binary directly off of the linux kernel.

• Task 4:

The basic python-based GUI has been created with a preliminary design to allow for a unified integration of local tools. However, an obstacle within this is creating an iframe section for local tools to be displayed within. This is a capability that none of the team was familiar with and therefore required research.

• Task 5:

In order to get the testbed set-up we needed to create a NixOS config that would allow for a reproducible OS environment. In addition, we needed IT to provide us with an ethernet connection within the IOT lab for the purposes of network tasking. One obstacle is that Nathan did not flash the router with OpenWRT, and until that is done, we will be limited to the stock router firmware for customization and network-based research.

• Task 6:

Menus designed to allow the user to access all the new features have been added. This includes a new artifact upload page with metadata fields. A new script upload menu including various upload types for various script forms. An environment menu to create default container environments with new tools. A run menu to begin analysis using scripts, environments, and artifacts. There is also a usage menu to view cloud usage.

• Task 7:

Cloud backend was massively expanded to allow the core functionality of analysis to be fully functional. Scripts can now be uploaded in various forms and then can be paired with uploaded artifacts for analysis. This will allow users to review and analyze files in bulk using the cloud platform to its full potential. The file upload system has also been reworked to allow for a much finer and powerful file upload system. Artifacts can now be uploaded in batches and are no longer limited to firmware. They can also be given various forms of metadata to help organize them for use in analysis. Cloud usage is also tracked per user.

Task 8:

Completed the end-to-end device and firmware management workflow across the web frontend and AWS cloud backend. Authenticated users can create and view IoT/medical devices and upload firmware associated with those devices. Device and firmware operations were integrated with AWS services including Cognito, API Gateway, Lambda, DynamoDB, and S3. The firmware upload workflow includes SHA-256 hashing, firmware upload reservation, presigned S3 uploads, upload completion verification, and retry handling for interrupted uploads. Firmware metadata and upload status can also be retrieved and displayed through the web interface. State-management and authorization protections were implemented using firmware and upload-attempt identifiers to prevent stale or conflicting retries and restrict access to resources owned by the authenticated user. During integration and testing, issues involving CORS and multiple CloudFront/domain origins were encountered. The team addressed this by standardizing the application around the primary domain and working toward redirecting alternate URLs to the canonical domain.

7. Discussion of contribution of each team member to Milestone 4:

Justin Bower:

- Maintained all email communications with advisor and communicated their comments, suggestions, and requests to the rest of the team
- Created, shared, and did the most work for every project document beside this one
- Read advisors research paper and planned I2C sensor integration in to the project using a bus pirate tool for analysis and logging
- Continued Adafruit bluetooth pcap research and created a demo for the next milestone presentation
- Completed/documented all the known CVEs for the wifi chipsets found in sem. 1

- Extracted first firmware binary from a wifi chipset from the open source linux driver with a reproducible workflow.
- Began the local GUI work in Python to unify tooling and passed it to Jordan directly as it took time for Nathan to provide him Github access
- Set up physical testbed along with Jordan and Ipule; moreover, finished it by wiping a lab computer and provided the team with a starter OS configuration file for setting up a reproducible environment with the necessary tooling
- Reverse engineered oximeter devices and found no documented CVEs.

Nathan Maloney:

- Created the plan for all backend and infrastructure changes
- Planned out the setup for new upload pipeline, script management, environment creation, analysis runs, and usage tracking
- Implemented all of the same features and created a web GUI for them
- Reworked firmware upload system to work for all upload types and expanded functionality to allow for batch uploads
- Managed the deployment of the cloud infrastructure and creation of the services that support the new features
- Created a demo of the new analysis system in use

Ipule Pipi:

- Completed the authenticated device management workflow for adding and viewing IoT/medical devices through the web dashboard.
- Integrated the device management workflow with AWS services, including API Gateway, Lambda, DynamoDB, and Cognito, allowing device records to be associated with authenticated users.
- Implemented the end-to-end firmware upload workflow, including firmware reservation, SHA-256 hashing, presigned S3 uploads, upload completion, and retry handling for interrupted upload attempts.
- Implemented the firmware viewing interface so uploaded firmware and associated metadata can be retrieved and displayed for registered devices.
- Improved firmware upload state management and retry behavior using firmware and upload-attempt identifiers to prevent stale or conflicting upload attempts.
- Worked on frontend/backend integration issues, including CORS and domain configuration problems affecting firmware uploads through different CloudFront/domain origins.
- Assisted Justin and Jordan in setting up the lab computer environment, including configuring Linux/NixOS and troubleshooting setup and configuration issues.
- Tested the web-based device and firmware workflow using actual firmware files and helped verify that the device creation and firmware upload pipeline worked end-to-end.

Jordan Chesley:

- Provided the flash drive containing the NixOS installer and worked with Justin and Ipule to configure the physical testbed's OS environment.
- Deconstructed oximeters for firmware extraction. (Have not identified CVEs yet.)
- Configured display passthrough within the Local GUI dev container so that GUI applications can be run within the dev container.
- Implemented script subprocesses to run tools through the Local GUI and print their output within the GUI window.

8. Plan for the next Milestone (Milestone 5 -)

Task	Justin	Nathan	Ipule	Jordan
DS18B20 I2C Sensor/Buspirate Research and Integration	Spearhead Buspirate and sensor research/integration	Assist with the research and integration	Assist with the research and integration	Assist with the research and integration
Containerized Deployments and Platform Security Testing	Work with Jordan to port network tools to the GUI as needed and write deployment scripts for their cloud VM setup.	Assist with the creation and deployment of various docker containers	Assist with the creation and deployment of various docker containers	Work with Justin to port network tools to the GUI as needed and write deployment scripts for their cloud VM setup.
Start-to-Finish Proof-of-Concept Demo	Verify integrations and create the demo	Create pipelines to allow for files to be easily used in each step of RE work	Create pipelines to allow for files to be easily used in each step of RE work	Create a firmware analysis / exploitation pipeline for extracted firmware
Project Management, Communication, and Documents	Manage project, documents, and communications	Assist with document creation	Assist with document creation	Assist with document creation

Network Traffic Analysis Integration (Wireshark/Tshark)	Create & integrate Wireshark / Tshark scripts with Local GUI	Assist with testing	Assist with testing	Create & integrate Wireshark / Tshark scripts with Local GUI
Firmware RE & Analysis (Ghidra workflows, entropy analysis, Binwalk/Foremost segmentation)	Create & integrate Ghidra analysis / file-carving scripts with Local GUI	Assist with testing	Assist with testing	Create & integrate Ghidra analysis / file-carving scripts with Local GUI
WiFi/Bluetooth Exploitation Script Development	Continue development of pre-made exploitation scripts for wifi/bluetooth exploitation	Assist with research, implementation, and testing	Assist with research, implementation, and testing	Create & integrate WiFi/Bluetooth analysis scripts with Local GUI
Local GUI Development	Assist Jordan with research, implementation, and testing	Create API to allow local GUI to communicate with cloud securely	Assist with API	Design UI/UX & integrate tools into application
Real time file analysis on cloud platform	Test security with Nathan by ensuring proper file parsing with no code injection	Create systems for users to be able to run jupyter notebooks for real time file analysis on the cloud	Create GUI that allows user to see the notebook and its results in real time	Test security with Nathan by ensuring proper file parsing with no code injection

9. Discussion of each planned task for Milestone 5:

DS18B20 I2C Sensor/Buspirate Research and Integration: Justin will lead the research and integration of the DS18B20 temperature sensor over I2C using a Buspirate for analysis and logging. This builds directly on the I2C planning already started and the oximeter RE work. Nathan, Ipule, and Jordan will assist with the research, implementation, and testing so the sensor can be used as a concrete demo target inside the larger testbed.

Containerized Deployments and Platform Security Testing: Justin and Jordan will work together to port the necessary network tools into the local GUI and write the deployment scripts needed for the cloud VM setup. Nathan and Ipule will handle creation and deployment of the Docker containers that isolate analysis work. Security testing of the containerized environments and the overall platform will be performed in parallel so that file parsing and tool execution cannot be abused for code injection.

Start-to-Finish Proof-of-Concept Demo: Justin will verify that the individual pieces integrate cleanly and will assemble the overall demo. Nathan and Ipule will create the pipelines that move files cleanly between each stage of RE work. Jordan will then build the firmware analysis and exploitation pipelines, which turn extracted firmware through to analyzed results usable for extrapolating novel ideas. The goal is a single, reproducible methodology that can be demonstrated end-to-end.

Project Management, Communication, and Documents: Justin will continue to manage the project schedule, all documentation, and communications with the advisors. Nathan, Ipule, and Jordan will assist with document creation so the workload is no longer concentrated on one person.

Network Traffic Analysis Integration (Wireshark/Tshark): Justin and Jordan will create the scripts that drive Wireshark/Tshark and integrate them into the local GUI so captures can be launched and results viewed without leaving the application. Nathan and Ipule will assist with testing against live traffic from the testbed.

Firmware RE & Analysis (Ghidra workflows, entropy analysis, Binwalk/Foremost segmentation): Justin and Jordan will develop the Ghidra analysis scripts and the file-carving/entropy scripts (Binwalk/Foremost) and wire them into the local GUI. Nathan and Ipule will assist with testing the resulting workflows on real extracted firmware.

WiFi/Bluetooth Exploitation Script Development: Justin will continue building the pre-made exploitation scripts for WiFi and Bluetooth targets. Nathan and Ipule will help with research, implementation, and testing. Jordan will create and integrate the corresponding analysis scripts into the local GUI so the scripts can be launched and their output captured inside the application.

Local GUI Development: Jordan will be responsible for the UI/UX design and the integration of the various tools into the application. Justin will assist with research, implementation, and testing. Nathan will build the secure API that lets the local GUI talk to the cloud backend; Ipule will assist with that API work. Remaining items include launch options, in-frame tool display, and reliable script launching.

Real-time file analysis on cloud platform: Nathan will create the systems that let users run Jupyter notebooks for live file analysis in the cloud. Ipule will build the GUI that streams the notebook and its results back to the user in real time. Justin and Jordan will work with Nathan on the security side, confirming that file parsing and execution paths cannot be used for code injection.

10. Date(s) of meeting(s) with Client during the current milestone:

9/11/26 @ 1:00 p.m., 9/25/26 @ 1:00 p.m.

11. Client feedback on the current milestone

- See Faculty Advisor Feedback below

12. Date(s) of meeting(s) with Faculty Advisor during the current milestone:

3/30/26 @ 2:00 p.m. and 4/17/26 @ 12:00 p.m.

13. Faculty Advisor feedback on each task for the current Milestone

- Task 1:

- Task 2:

- Task 3:

- Task 4:

- Task 5:

Faculty Advisor Signature: _____ Date: _____